

Homework (Carolina Reaper)

- Q1.** A music producer wants to ensure that the frequency of a song is between $70 \cdot 2$ and $70 \cdot 3$ Hz. Which absolute value inequality represents this situation?
- (A) $|x - 70| \leq 35$
(B) $|x - 105| \leq 35$
(C) $|x - 70| \geq 35$
(D) $|x - 105| \geq 35$
(E) $|x - 70| \leq 70$
- Q2.** The width of a painting is x inches. If the width must be within 10 inches of 50 inches, which inequality represents this?
- (A) $|x - 50| \leq 10$
(B) $|x - 50| \geq 10$
(C) $|x - 40| \leq 10$
(D) $|x - 60| \leq 10$
(E) $|x - 50| = 10$
- Q3.** A rhythm pattern has a duration of x seconds. If the duration must be within 2 seconds of 15 seconds, which inequality represents this situation?
- (A) $|x - 15| \leq 2$
(B) $|x - 15| \geq 2$
(C) $|x - 13| \leq 2$
(D) $|x - 17| \leq 2$
(E) $|x - 15| = 2$
- Q4.** The number of attendees at a concert is x . If the number of attendees must be within 50 of 200, which inequality represents this situation?
- (A) $|x - 200| \leq 50$
(B) $|x - 200| \geq 50$
(C) $|x - 150| \leq 50$
(D) $|x - 250| \leq 50$
(E) $|x - 200| = 50$
- Q5.** A sound engineer wants to ensure that the sound level is between $60 \cdot 2$ and $60 \cdot 3$ decibels. Which absolute value inequality represents this situation?
- (A) $|x - 60| \leq 30$
(B) $|x - 120| \leq 30$
(C) $|x - 60| \geq 30$
(D) $|x - 120| \geq 30$
(E) $|x - 60| \leq 60$
- Q6.** The length of a canvas is x feet. If the length must be within 2 feet of 10 feet, which inequality represents this?
- (A) $|x - 10| \leq 2$
(B) $|x - 10| \geq 2$
(C) $|x - 8| \leq 2$
(D) $|x - 12| \leq 2$
(E) $|x - 10| = 2$
- Q7.** A music teacher wants to ensure that the tempo of a song is between $100 \cdot 2$ and $100 \cdot 3$ beats per minute. Which absolute value inequality represents this situation?
- (A) $|x - 100| \leq 50$
(B) $|x - 200| \leq 50$
(C) $|x - 100| \geq 50$
(D) $|x - 200| \geq 50$
(E) $|x - 100| \leq 100$
- Q8.** The cost of painting a mural is x dollars. If the cost must be within 100 of 500, which inequality represents this situation?
- (A) $|x - 500| \leq 100$
(B) $|x - 500| \geq 100$
(C) $|x - 400| \leq 100$
(D) $|x - 600| \leq 100$
(E) $|x - 500| = 100$

Open Ended Questions (FRQ)

- Q9.** A musician is setting up a sound system for a concert. The sound level must be within 20 decibels of 80 decibels. Write an absolute value inequality to represent this situation and graph the solution set on a number line.
- Q10.** An artist is creating a sculpture with a height of x inches. The height must be within 5 inches of 30 inches. Write an absolute value inequality to represent this situation and express the solution in interval notation.

Chef's Tips

- Tip 1: Ensure students understand the concept of absolute value inequalities and their applications.
- Tip 2: Encourage students to use graphing to visualize the solution sets.
- Tip 3: Remind students to check their work and provide clear explanations for their answers.